

## Lab 01: The Rental Refund Crisis

Query Query History

```
1 SELECT rental_id , customer_id FROM rental LIMIT 1;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

|   | rental_id<br>[PK] integer | customer_id<br>smallint |
|---|---------------------------|-------------------------|
| 1 | 2                         | 459                     |

### Exercise 1:

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Servers (1)
  - PostgreSQL 18
    - Databases (2)
      - dvdrental
        - Casts
        - Catalogs
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      - postgres
      - Login/Group Roles
      - Tablespaces

dvdrental/postgres@PostgreSQL 18

Query Query History

```
1 SELECT rental_id , customer_id FROM rental LIMIT 1;
2
3 BEGIN;
4 -- Step 1: Mark as returned
5 UPDATE rental SET return_date = NOW() WHERE rental_id = 2;
6
7 -- Step 2: Intentional Error
8 INSERT INTO payment(customer_id, staff_id, rental_id, amount, payment_date)
9 VALUES(459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
10
11 COMMIT;
12
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Total rows: 1 Query complete 00:00:00.088 CRLF Ln 11, Col 8

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File Object Tools Edit View Window Help

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dvdrental/postgres@PostgreSQL 18

Query Query History

```
11 COMMIT;  
12  
13 SELECT return_date FROM rental WHERE rental_id=2;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Total rows: 1 Query complete 00:00:00.096 CRLF Ln 13, Col 50

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File Object Tools Edit View Window Help

Object Explorer

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dvdrental/postgres@PostgreSQL 18

Query Query History

```
13 SELECT return_date FROM rental WHERE rental_id=2;  
14  
15 ROLLBACK;
```

Data Output Messages Notifications

|   | return_date                 |
|---|-----------------------------|
|   | timestamp without time zone |
| 1 | 2005-05-28 19:40:33         |

Showing rows: 1 to 1 Page No: 1 of 1

Total rows: 1 Query complete 00:00:00.083 CRLF Ln 14, Col 1

Savepoints: Partial Rollbacks

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File Object Tools Edit View Window Help

Object Explorer

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dvdrental/postgres@PostgreSQL 18

Query

```

18
19 BEGIN;
20 INSERT INTO customer(first_name, last_name, email , address_id, store_id)
21 VALUES ('AI', 'Student', 'ai@edu.vn', 1, 1);
22
23 SAVEPOINT user_created;
24
25 INSERT INTO special_offers(customer_id, code) VALUES (9999, 'FREE!');
26 ROLLBACK TO user_created;
27
28 COMMIT;

```

Data Output Messages Notifications

ERROR: relation "special\_offers" does not exist  
 LINE 7: INSERT INTO special\_offers (customer\_id, code) VALUES (9999...

SQL state: 42P01  
 Character: 171

Total rows: 1 Query complete 00:00:00.108 CRLF Ln 28, Col 8

## Exercise 2:

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File Object Tools Edit View Window Help

Object Explorer

- Servers (1)
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dvdrental/postgres@PostgreSQL 18

Query

```

19 BEGIN;
20 INSERT INTO staff(first_name, last_name, email , address_id, store_id, username, password)
21 VALUES ('Thu', 'Nguyen', 'anhthu@gmail.vn', 1, 1, 'anhthu_lab', 12345678);
22
23 SAVEPOINT staff_created;
24
25 UPDATE staff
26 SET store_id = 999
27 WHERE username = 'anhthu_lab';
28
29 COMMIT;

```

Data Output Messages Notifications

SAVEPOINT

Query returned successfully in 79 msec.

Total rows: Query complete 00:00:00.079 CRLF Ln 23, Col 26

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File Object Tools Edit View Window Help

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dvdrental/postgres@PostgreSQL 18

Query Query History

```
16
17
18
19 BEGIN;
20 INSERT INTO staff(first_name, last_name, email , address_id, store_id, username, password)
21 VALUES ('Thu', 'Nguyen', 'anhthu@gmail.vn', 1, 1, 'anhthu_lab', 12345678);
22
23 SAVEPOINT staff_created;
24
25 UPDATE staff
26 SET store_id = 999
27 WHERE username = 'anhthu_lab';
28
```

Data Output Messages Notifications

ERROR: insert or update on table "staff" violates foreign key constraint "staff\_store\_id\_fkey"  
Key (store\_id)=(999) is not present in table "store".

SQL state: 23503  
Detail: Key (store\_id)=(999) is not present in table "store".

Total rows: Query complete 00:00:00.102 CRLF Ln 27, Col 32

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File Object Tools Edit View Window Help

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dvdrental/postgres@PostgreSQL 18

Query Query History

```
23 SAVEPOINT staff_created;
24
25 UPDATE staff
26 SET store_id = 999
27 WHERE username = 'anhthu_lab';
28
29 ROLLBACK TO SAVEPOINT staff_created;
30
31 UPDATE staff
32 SET store_id = 1
33 WHERE username = 'anhthu_lab';
34
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 81 msec.

✓ Query returned successfully in 81 msec. ✕

Total rows: Query complete 00:00:00.081 CRLF Ln 29, Col 38

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File Object Tools Edit View Window Help

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dvdrental/postgres@PostgreSQL 18\*

Query Query History

```
24
25 UPDATE staff
26 SET store_id = 999
27 WHERE username = 'anhthu_lab';
28
29 ROLLBACK TO SAVEPOINT staff_created;
30
31 UPDATE staff
32 SET store_id = 1
33 WHERE username = 'anhthu_lab';
34
35 COMMIT;
36
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 79 msec.

Total rows: Query complete 00:00:00.079 CRLF Ln 31, Col 2